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Probability and Statistics Lab (DA-2)

1. **Question 1:** A transport analyst wants to study the relationship between daily advertising expenditure (X) in thousand rupees and number of passengers (Y) using a new ride-sharing service over 8 days.

Day	Advertising Expenditure (X)	Number of Passengers (Y)
1	5	120
2	8	150
3	6	135
4	10	180
5	7	145
6	12	200
7	9	165
8	11	190

Calculate Karl Pearson's coefficient of correlation between advertising expenditure and number of passengers.

CODE:-

```
DA1.R x
Source on Save
Run
Source

1 # -----
2 # 1. Enter the Data
3 # -----
4
5 # Advertising Expenditure (in thousand rupees)
6 X <- c(5, 8, 6, 10, 7, 12, 9, 11)
7
8 # Number of Passengers
9 Y <- c(120, 150, 135, 180, 145, 200, 165, 190)
10
11 # -----
12 # 2. Descriptive Statistics
13 # -----
14
15 mean_X <- mean(X)
16 mean_Y <- mean(Y)
17
18 sd_X <- sd(X)
19 sd_Y <- sd(Y)
20
21 cat("Mean of X:", mean_X, "\n")
22 cat("Mean of Y:", mean_Y, "\n")
23 cat("Standard Deviation of X:", sd_X, "\n")
24 cat("Standard Deviation of Y:", sd_Y, "\n\n")
25
26 # -----
27 # 3. Karl Pearson Correlation
28 # -----
29
30 correlation <- cor(X, Y, method = "pearson")
31 cat("Karl Pearson Correlation (r):", correlation, "\n\n")
32
33 # -----
34 # 4. Correlation Test
35 # -----
36
37 cor_test <- cor.test(X, Y, method = "pearson")
38 print(cor_test)
39

80:1 (Untitled) R Script
```

CONSOLE:-

```
Console Terminal x Background Jobs x
R - R 4.5.2 - ~/new da/
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> # -----
> # End of Code
> # -----
> # -----
> # End of Code
> # -----
> # -----
> # End of Code
> # -----
> source("~/new da/aryan/DA1.R")
Mean of X: 8.5
Mean of Y: 160.625
Standard Deviation of X: 2.44949
Standard Deviation of Y: 27.95883

Karl Pearson Correlation (r): 0.9960489

Pearson's product-moment correlation

data: X and Y
t = 27.473, df = 6, p-value = 1.538e-07
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.9774065 0.9993144
sample estimates:
      cor
0.9960489

Manual Calculation of r: 0.9960489

> |
```

OUTPUT:-

Environment

History

Connections

Tutorial

Import Dataset

20 MiB

List

R

Global Environment

Data

cor_test	List of 9	
model	List of 12	

Values

correlation	0.996048869380638
dx	num [1:8] -3.5 -0.5 -2.5 1.5 -1.5 3.5 0.5 2.5
dy	num [1:8] -40.6 -10.6 -25.6 19.4 -15.6 ...
mean_X	8.5
mean_Y	160.625
r_manual	0.996048869380638

Files

Plots

Packages

Help

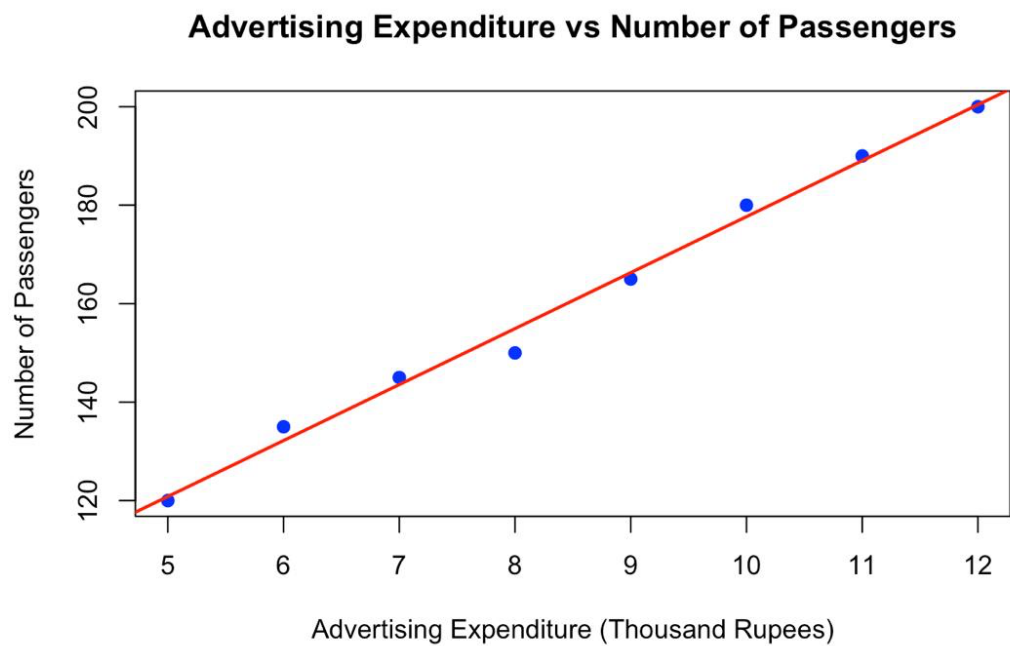
Viewer

Presentation

Zoom

Export

Publish



Que 1 (code part):-

1. Enter the Data

Advertising Expenditure (in thousand rupees)

```
X <- c(5, 8, 6, 10, 7, 12, 9, 11)
```

Number of Passengers

```
Y <- c(120, 150, 135, 180, 145, 200, 165, 190)
```

2. Descriptive Statistics

```
mean_X <- mean(X) mean_Y <- mean(Y)
```

```
sd_X <- sd(X) sd_Y <- sd(Y)
```

```
cat("Mean of X:", mean_X, "\n") cat("Mean of Y:", mean_Y, "\n") cat("Standard Deviation of X:", sd_X, "\n") cat("Standard Deviation of Y:", sd_Y, "\n\n")
```

3. Karl Pearson Correlation

```
correlation <- cor(X, Y, method = "pearson") cat("Karl Pearson Correlation (r):", correlation,
"\n\n")
```

4. Correlation Test

```
cor_test <- cor.test(X, Y, method = "pearson") print(cor_test)
```

5. Manual Calculation of r

Deviations

```
dx <- X - mean_X dy <- Y - mean_Y
```

Formula components

```
sum_dxdy <- sum(dx * dy) sum_dx2 <- sum(dx^2) sum_dy2 <- sum(dy^2)
```

```
r_manual <- sum_dxdy / sqrt(sum_dx2 * sum_dy2)
```

```
cat("\nManual Calculation of r:", r_manual, "\n\n")
```

6. Simple Linear Regression

```
model <- lm(Y ~ X) summary(model)
```

7. Scatter Plot with Regression Line

```
plot(X, Y, main = "Advertising Expenditure vs Number of Passengers", xlab = "Advertising  
Expenditure (Thousand Rupees)", ylab = "Number of Passengers", pch = 19, col = "blue")
```

```
abline(model, col = "red", lwd = 2)
```

End of Code

QUE-2

2. A researcher ranked 10 employees based on Performance Score (X) and Leadership Score (Y) as shown below:

Employee	X	Y
1	85	88
2	78	75
3	92	90
4	70	72
5	88	85
6	75	78
7	80	82
8	90	92
9	68	70
10	82	80

Compute Spearman's rank correlation coefficient.

CODE:-


```
Untitled1* x
Source on Save
Run
Source

1 # -----
2 # 1. Enter the Data
3 # -----
4
5 # Performance Score (X)
6 X <- c(85, 78, 92, 70, 88, 75, 80, 90, 68, 82)
7
8 # Leadership Score (Y)
9 Y <- c(88, 75, 90, 72, 85, 78, 82, 92, 70, 80)
10
11 # -----
12 # 2. Spearman Rank Correlation
13 # -----
14
15 spearman_corr <- cor(X, Y, method = "spearman")
16
17 cat("Spearman's Rank Correlation (rho):", spearman_corr, "\n\n")
18
19 # -----
20 # 3. Significance Test
21 # -----
22
23 spearman_test <- cor.test(X, Y, method = "spearman")
24
25 print(spearman_test)
26
27 # -----
28 # 4. Manual Calculation Method
29 # -----
30
31 # Rank the data
32 rank_X <- rank(X)
33 rank_Y <- rank(Y)
34
35 # Difference in ranks
36 d <- rank_X - rank_Y
37 d2 <- d^2
38
39 # Apply Spearman formula
48:1 # (Untitled)
R Script
```

CONSOLE:-

```
Source
Console Terminal x Background Jobs x
R 4.5.2 · ~/dhcnsdfb/
> spearman_test <- cor.test(X, Y, method = "spearman")
>
> print(spearman_test)

Spearman's rank correlation rho

data: X and Y
S = 8, p-value < 2.2e-16
alternative hypothesis: true rho is not equal to 0
sample estimates:
rho
0.9515152

>
> # -----
> # 4. Manual Calculation Method
> # -----
>
> # Rank the data
> rank_X <- rank(X)
> rank_Y <- rank(Y)
>
> # Difference in ranks
> d <- rank_X - rank_Y
> d2 <- d^2
>
> # Apply Spearman formula
> n <- length(X)
> rho_manual <- 1 - (6 * sum(d2)) / (n * (n^2 - 1))
>
> cat("\nManual Spearman's rho:", rho_manual, "\n")

Manual Spearman's rho: 0.9515152
>
> # -----
> # End of Code
> # -----
>
```

OUTPUT:-

Environment

History

Connections

Tutorial

Import Dataset

71 MiB

List

R

Global Environment

Data

spearman_test

List of 8

Values

d	num [1:10] -1 1 1 0 1 -1 -1 -1 0 1
d2	num [1:10] 1 1 1 0 1 1 1 1 0 1
n	10L
rank_X	num [1:10] 7 4 10 2 8 3 5 9 1 6
rank_Y	num [1:10] 8 3 9 2 7 4 6 10 1 5
rho_manual	0.951515151515152
spearman_corr	0.951515151515151
X	num [1:10] 85 78 92 70 88 75 80 90 68 82
Y	num [1:10] 88 75 90 72 85 78 82 92 70 80

Files	Plots	Packages	Help	Viewer	Presentation
New Folder New File Delete Rename More					
Home > dhcnsdfb					
..			Name		Size
					Modified

Que -2(code part):-

1. Enter the Data

Performance Score (X)

```
X <- c(85, 78, 92, 70, 88, 75, 80, 90, 68, 82)
```

Leadership Score (Y)

```
Y <- c(88, 75, 90, 72, 85, 78, 82, 92, 70, 80)
```

2. Spearman Rank Correlation

```
spearman_corr <- cor(X, Y, method = "spearman")
```

```
cat("Spearman's Rank Correlation (rho):", spearman_corr, "\n\n")
```

3. Significance Test

```
spearman_test <- cor.test(X, Y, method = "spearman")  
print(spearman_test)
```

4. Manual Calculation Method

Rank the data

```
rank_X <- rank(X) rank_Y <- rank(Y)
```

Difference in ranks

```
d <- rank_X - rank_Y d2 <- d^2
```

Apply Spearman formula

```
n <- length(X) rho_manual <- 1 - (6 * sum(d2)) / (n * (n^2 - 1))  
cat("\nManual Spearman's rho:", rho_manual, "\n")
```

End of Code

QUESTION 3:-

3. A logistics company wants to examine how daily fuel consumption depends on the total distance travelled by delivery vehicles.

The following data were collected for 15 working days:

Day	Distance Travelled (km) (X)	Fuel Consumption (litres) (Y)
1	120	18
2	150	22
3	180	26
4	200	29
5	160	24
6	170	25
7	140	21
8	210	31
9	190	27
10	220	33
11	155	23
12	165	24
13	175	26
14	205	30
15	195	28

Fit a simple linear regression line of fuel consumption (Y) on distance travelled (X).
Estimate the fuel consumption when the distance travelled is 185 km.

CODE:-

```
Untitled.R* x
Source on Save
Run
Source

1 # -----
2 # 1. Enter the Data
3 # -----
4
5 # Distance Travelled (km)
6 X <- c(120,150,180,200,160,170,140,210,190,220,
7        155,165,175,205,195)
8
9 # Fuel Consumption (litres)
10 Y <- c(18,22,26,29,24,25,21,31,27,33,
11        23,24,26,30,28)
12
13 # -----
14 # 2. Fit Linear Regression Model
15 # -----
16
17 model <- lm(Y ~ X)
18
19 # Display model summary
20 summary(model)
21
22 # -----
23 # 3. Regression Equation
24 # -----
25
26 intercept <- coef(model)[1]
27 slope <- coef(model)[2]
28
29 cat("Regression Equation:\n")
30 cat("Y =", round(intercept,4), "+", round(slope,4), "X\n\n")
31
32 # -----
33 # 4. Prediction for X = 185 km
34 # -----
35
36 predicted_value <- predict(model, newdata = data.frame(X = 185))
37
38 cat("Estimated fuel consumption for 185 km:",
39     round(predicted_value,4), "litres\n")
43:34 ## (Untitled) R Script
```

CONSOLE:-


```
Console Terminal × Background Jobs ×
R 4.5.2 · ~/dhcnsdfb/

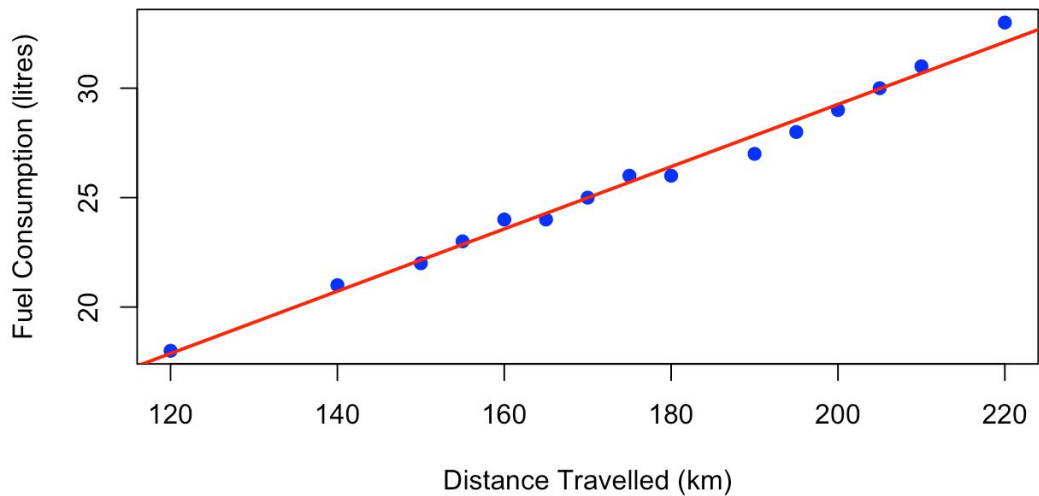
>
> # -----
> # 3. Regression Equation
> # -----
>
> intercept <- coef(model)[1]
> slope <- coef(model)[2]
>
> cat("Regression Equation:\n")
Regression Equation:
> cat("Y =", round(intercept,4), "+", round(slope,4), "X\n\n")
Y = 0.7941 + 0.1423 X

>
> # -----
> # 4. Prediction for X = 185 km
> # -----
>
> predicted_value <- predict(model, newdata = data.frame(X = 185))
>
> cat("Estimated fuel consumption for 185 km:",
+     round(predicted_value,4), "litres\n")
Estimated fuel consumption for 185 km: 27.1286 litres
>
> # -----
> # 5. Scatter Plot with Regression Line
> # -----
>
> plot(X, Y,
+      main = "Fuel Consumption vs Distance Travelled",
+      xlab = "Distance Travelled (km)",
+      ylab = "Fuel Consumption (litres)",
+      pch = 19,
+      col = "blue")
>
> abline(model, col = "red", lwd = 2)
>
> |
```

OUTPUT:-

R	Global Environment	
Data		
model	List of 12	
spearman_test	List of 8	
Values		
d	num [1:10] -1 1 1 0 1 -1 -1 -1 0 1	
d2	num [1:10] 1 1 1 0 1 1 1 1 0 1	
intercept	Named num 0.794	
n	10L	
predicted_value	Named num 27.1	
rank_X	num [1:10] 7 4 10 2 8 3 5 9 1 6	
rank_Y	num [1:10] 8 3 9 2 7 4 6 10 1 5	
rho_manual	0.951515151515152	
rho	Named num 0.942	
Files	Plots	Packages Help Viewer Presentation
Zoom	Export	Publish

Fuel Consumption vs Distance Travelled



Que-3(code part):-

1. Enter the Data

Distance Travelled (km)

```
X <- c(120,150,180,200,160,170,140,210,190,220, 155,165,175,205,195)
```

Fuel Consumption (litres)

```
Y <- c(18,22,26,29,24,25,21,31,27,33, 23,24,26,30,28)
```

2. Fit Linear Regression Model

```
model <- lm(Y ~ X)
```

Display model summary

```
summary(model)
```

3. Regression Equation

```
intercept <- coef(model)[1] slope <- coef(model)[2]
cat("Regression Equation:\n") cat("Y =", round(intercept,4), "+", round(slope,4), "X\n\n")
```

4. Prediction for X = 185 km

```
predicted_value <- predict(model, newdata = data.frame(X = 185))
cat("Estimated fuel consumption for 185 km:", round(predicted_value,4), "litres\n")
```

5. Scatter Plot with Regression Line

```
plot(X, Y, main = "Fuel Consumption vs Distance Travelled", xlab = "Distance Travelled
(km)", ylab = "Fuel Consumption (litres)", pch = 19, col = "blue")

abline(model, col = "red", lwd = 2)
```

Question –4

4. A retail company wants to analyse how monthly sales revenue depends on advertising expenditure and store footfall. For 15 randomly selected months, the following data were recorded:

Month	Advertising Expenditure (₹ '000) (X_1)	Average Daily Footfall (X_2)	Sales Revenue (₹ '000) (Y)
1	25	180	410
2	30	200	455
3	28	190	435
4	35	220	500
5	32	210	480

6	27	185	420
7	40	240	550
8	38	230	530
9	34	215	490
10	45	260	590
11	33	205	470
12	29	195	440
13	42	250	570
14	36	225	510
15	31	200	460

Fit a multiple linear regression model of Sales Revenue (Y) on Advertising Expenditure (X_1) and Footfall (X_2). Estimate the expected sales revenue when advertising expenditure is ₹37,000 and footfall is 235.

Code:-

```
Untitled.R* x
Source on Save
Run
Source

1 # -----
2 # 1. Enter the Data
3 # -----
4
5 # Advertising Expenditure (₹ '000)
6 X1 <- c(25,30,28,35,32,27,40,38,34,45,
7         33,29,42,36,31)
8
9 # Average Daily Footfall
10 X2 <- c(180,200,190,220,210,185,240,230,215,260,
11         205,195,250,225,200)
12
13 # Sales Revenue (₹ '000)
14 Y <- c(410,455,435,500,480,420,550,530,490,590,
15        470,440,570,510,460)
16
17 # -----
18 # 2. Fit Multiple Linear Regression
19 # -----
20
21 model <- lm(Y ~ X1 + X2)
22
23 summary(model)
24
25 # -----
26 # 3. Regression Equation
27 # -----
28
29 coef(model)
30
31 cat("\nRegression Equation:\n")
32 cat("Y =",
33     round(coef(model)[1],4), "+",
34     round(coef(model)[2],4), "X1 +",
35     round(coef(model)[3],4), "X2\n")
36
```

Console:-

```

Console | Terminal x | Background Jobs x
R 4.5.2 · ~/dhcnsdfb/
> # 5. Scatter Plot with Regression Line
> # -----
> > plot(X, Y,
+       main = "Fuel Consumption vs Distance Travelled",
+       xlab = "Distance Travelled (km)",
+       ylab = "Fuel Consumption (litres)",
+       pch = 19,
+       col = "blue")
> > abline(model, col = "red", lwd = 2)
> > >
> source("~/dhcnsdfb/Untitled.R")

Regression Equation:
Y = 40.1093 + 2.3094 X1 + 1.7292 X2

Estimated Sales Revenue (₹ '000): 531.9211
>

```

Output:-

Environment	History	Connections	Tutorial
Import Dataset 112 MiB			
R	Global Environment		
Data			
model	List of 12		
new_data	1 obs. of 2 variables		
spearman_test	List of 8		
Values			
d	num [1:10] -1 1 1 0 1 -1 -1 -1 0 1		
d2	num [1:10] 1 1 1 0 1 1 1 1 0 1		
intercept	Named num 0.794		
n	10L		
predicted_sales	Named num 532		
predicted_value	Named num 27.1		
rank_X	num [1:10] 7 4 10 2 8 3 5 9 1 6		
rank_Y	num [1:10] 8 3 9 2 7 4 6 10 1 5		
rho_manual	0.951515151515152		
slope	Named num 0.142		
spearman_corr	0.951515151515151		
X	num [1:15] 120 150 180 200 160 170 140 210 190 220 ...		
X1	num [1:15] 25 30 28 35 32 27 40 38 34 45 ...		
X2	num [1:15] 180 200 190 220 210 185 240 230 215 260 ...		
Y	num [1:15] 41 num [1:15] 180 200 190 220 210 185 240 230 215 260 ...		

Question-4(code)

1. Enter the Data

Advertising Expenditure (₹ '000)

```
X1 <- c(25,30,28,35,32,27,40,38,34,45, 33,29,42,36,31)
```

Average Daily Footfall

```
X2 <- c(180,200,190,220,210,185,240,230,215,260, 205,195,250,225,200)
```

Sales Revenue (₹ '000)

```
Y <- c(410,455,435,500,480,420,550,530,490,590, 470,440,570,510,460)
```

2. Fit Multiple Linear Regression

```
model <- lm(Y ~ X1 + X2)
```

```
summary(model)
```

3. Regression Equation

```
coef(model)
```

```
cat("\nRegression Equation:\n") cat("Y =", round(coef(model)[1],4), "+",  
round(coef(model)[2],4),"X1 +", round(coef(model)[3],4),"X2\n")
```

4. Prediction

Advertising = 37 (₹37,000)

Footfall = 235

```
new_data <- data.frame(X1 = 37, X2 = 235)
```

```
predicted_sales <- predict(model, new_data)
```

```
cat("\nEstimated Sales Revenue (₹ '000):", round(predicted_sales,4), "\n")
```